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Deep Learning Techniques in Cyclone Detection with Cyclone Eye Localization Based on Satellite Images

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Abstract. Contemporary progress in satellite imaging and computer vision technology has made it feasible to make automatic weather predictions accurately. To reduce environmental damages and limit the risk of human lives, we have introduced a system to predict cyclones from satellite imagery. The eye of a powerful cyclone inherits a lot of information about its behavior, indicating that detecting the cyclone’s center rotating area is also an important task; we have constructed a Deep Learning object detection system to locate a cyclone’s eye. Several models using Machine Learning and Deep Learning approaches have produced a cyclone prediction accuracy, ranging from 86% to 95% or better, and cyclone eye detection accuracy over 87%. Our experiment’s results have been impressive, highlighting that it can outperform conventional approaches and existing works.

Keywords: Satellite Imagery · Remote Sensing · Machine Learning · Deep Learning · Cyclone Detection · Cyclone Eye Localization

1 Introduction

A cyclone is a natural catastrophe that occurs when a low-pressure circular spinning area arises over tropical waters, causing significant damage. It can also be called a ‘hurricane’, ‘typhoon’, etc., and its center rotation can be in the clockwise or anti-clockwise direction. It has caused countless damages all across the world due to its destructive characteristics like massive flooding, strong winds, and heavy downpours. Cyclones such as ‘Sidr’, ‘Aila’, ‘Amphan’, have destroyed numerous lives in Bangladesh. Thousands of people died in the Bay of Bengal area in 1970 as a result of the disastrous storm surge known as ‘Bhola’. As a result, cyclone prediction methods are becoming increasingly important.

To provide reliable results, it is important to understand the cloud patterns [1]. Forecasters have utilized satellite data to provide early warnings for cyclone formation; many of the outcomes were determined by human judgment, which is an ineffective procedure in today’s world. Researchers have used Machine

Learning and Deep Learning methods to overcome some of the constraints in weather forecasting. Nonetheless, additional study is required to produce a more dependable system.

Weather forecasting issues with Machine Learning algorithms have been reported in studies [2],[3]. Because of data availability, Deep Learning methods have been brought into the Remote Sensing field. It has achieved incredible performance in object recognition and classification with the help of graphical processing units (GPUs). Such models need processing capacity, which powerful GPUs deliver. Open-source Deep Learning software frameworks like TensorFlow and Keras have made the work simple. Research works on satellite image classification [4],[5] and Deep Learning algorithms in disaster detection [6] have given us the motivation to work in this field. To detect the cyclone and cyclone's eye in satellite images, we propose a Deep Learning based framework with multiple models. There are three major components to our proposed methodology: 1) Image Preprocessing, 2) Cyclone Detection, and 3) Cyclone Eye Localization.

We have worked with several Deep Learning models and achieved accurate cyclone identification. It is also necessary to locate the center of a cyclone. To address this issue, we have implemented an object-detecting system. This stage anticipates the central rotating region with a bounding box. A version of the You Only Look Once (YOLO) model, which has been trained in Google Colab with an Nvidia Tesla GPU returned the object localization result. Experiments have been performed on a series of satellite images obtained from the web and Kaggle competition. When compared to multiple models, our Deep Learning-based solution produced a higher recognition rate.

2 Related Works

Several methodologies for dealing with satellite images have lately been introduced. Traditional methodologies were introduced by some researchers, while Deep Learning and Machine Learning techniques were presented more recently. We'll go through some of the prior research that has been done in this field.

Matsuoka et al. detected tropical cyclones and their precursors using ensemble CNN, having data of 50,000 for cyclones and 500,000 for non-cyclones, they attained a maximum accuracy of 91.26 % [7]. Kim et al. used three distinct Machine Learning models (SVM, DT, and RF) to predict cyclone formation, with a hit rate ranging from 94 to 96 % [8]. Deep Learning models (CNN, MobileNet, NasNet, Xception) were used to classify cyclone images, and a linear regression model to forecast the future direction [9] with a minimum of 90% accuracy for classification and 84% for path prediction. Panangadan et al. developed a Kalman filter-based tracker that presented the results from five satellite sources, as well as an automated graph-based approach to locate cyclone's eye from satellite infrared images [10]. Wang et al. presented a two-step Deep Learning object detection approach 'OSIP', and 'IP' for detecting the precise center of a cyclone [11].

Many forecasters were aided by Dvorak’s invention in observing cloud characteristics changes in geostationary satellite images and estimating cyclone intensity [12],[13]. However, determining intensity in distinct cloud basins has proven to be challenging. Chen et al. proposed a modified CNN model for cyclone intensity estimation from satellite infrared and microwave data with an RMSE of 8.39 kt [14]. For cyclone track predictions, Kovordányi et al. presented a multi-layered Artificial Neural Network and obtained a 95% accuracy. However, they didn’t use a wide range of datasets [15]. Kussul et al. proposed an ensemble CNN architecture for the classification of land cover and crop type from Landsat-8 and Sentinel-1 satellite data and acquired 85% accuracy [16]. Deep neural network architecture such as VGG13 was applied to detect floods from the satellite images [17].

Additionally, our suggested approach can outperform many previous efforts and overcome their drawbacks. In the proposed method of Matsuoka et al., their classifiers can produce model-specific biases [7], where we have applied cross-validation techniques to prevent our models from bias and overfitting. Our method can accurately locate multiple cyclone eyes in a single image, where we have found the proposed deep learning algorithm of Shakya et al. unable to detect when images contain more than one cyclone. And their performance of classification can affect the cyclone eye localization [9]. We have performed the cyclone detection and eye localization task separately so that classification performance can’t influence the localization performance. We also don’t have to rely on expert human intuition to accomplish our tasks. We have also used some hybrid architecture of the Machine Learning and Deep Learning models, although some researchers have just used traditional Machine Learning models, and the results of our models are notable. Kovordányi et al. used relatively very low-resolution input images and found that distinguishing cyclone from the neighboring cumulonimbus clouds is difficult [15], in our case we have used much better resolution input images and also applied different preprocessing techniques on our images before training the models which can overcome the problem of distinguishing cyclones from the neighbor clouds. We have used several models, the classification accuracy is higher than most of the earlier works and for eye localization, we have used an advanced version of You Only Look Once (YOLO) that generated faster and accurate results.

3 Methodology

3.1 Dataset Description

The satellite images of the cyclone have been collected from Kaggle [18]. For the non-cyclone satellite images, some of them we have collected from the web [19], which contains cloud masks from the 2018 Level-1C Sentinel-2 archive. In addition, these images contained 11 categories of surface (dense tree cover, any snow or ice covering either land or water, farmlands or pastures, coastlines, somewhat sparsely vegetated areas, extended bodies of water, etc.), 7 categories of cloud (cumulus, cirrus, haze/fog, ice-clouds, etc.). We have also gathered some

images from Kaggle [20], which included Sentinel-2 satellite images of water bodies. We have created a new dataset using these images.

There are two parts to our dataset: training and testing for both detection and localization. We double-checked that no images were duplicated. For the detection purpose, the dataset contains 1460 images overall, with 837 images having cyclones and 623 images without a cyclone. The train set has 1151 images, accounting for about 80% of the whole dataset, whereas the test set has 309 images, accounting for slightly over 20% of the total data. Cyclone images have been categorized as positive, whereas non-cyclone images as negative. For localization, we have used the same 837 cyclone images. The train set contains 649 images and the test set contains 188 images.

3.2 Data Preprocessing

Preprocessing techniques have been used to clean image data for the model’s input. The images we gathered required to be preprocessed because there was the possibility of smaller cloud formations, shear, and center density overcast, as well as noise [21]. It has also decreased models training time and increased execution speed. Here are some preprocessing techniques we have applied to our images:

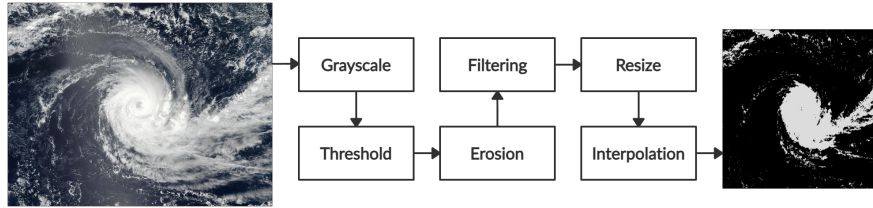


Fig. 1. Preprocessing Steps and The Outcome.

Images have been converted to grayscale. Working with grayscale images, our model will only need to keep track of one matrix per image, which reduces complexity and speeds up performance. Binary thresholding has been used to distinguish an object from the background or, in other words, it allows us to work with the areas of an image we are interested in. By applying thresholding, the cyclone’s eye and the surrounding clouds became more visible for training our models. We have used a size 2 erosion, which reduced the size of objects and eliminated minor irregularities, the output pixel’s value is the least of all the pixels in the neighborhood. Our dataset includes images of different sizes. Therefore, all the images have been resized to 300×300 pixels. The INTER_AREA interpolation method has also been used for enhancing the quality of an image. Figure 1, shows an example of an original and preprocessed image for cyclone detection.

For cyclone's eye localization, we have manually annotated the cyclone images. Using the 'Labellmg' tool, we took the images and annotated the bounding box.

3.3 Models for Detecting Cyclone

Convolutional Neural Network (CNN) : CNN can be thought of as Artificial Neural Networks that can recognize patterns. In our model, we have stacked the layers and formed an architecture as shown in Figure 2. The first layer consists of a 'convolutional' layer with 'Rectified Linear Unit' (ReLU); we have used 16 convolutional filters of size (3×3) , a stride of size 2. Then a 'max-pooling' layer with a pool and a stride of size 2 was added. After that, a second convolutional layer with 32 filters was implemented, followed by a max-pooling layer. Similarly, we have added two more convolutional and pooling layers. After these layers, the output was flattened to produce a one-dimensional vector for fully connected layers [6]. We have used two fully connected layers with the activation function ReLU. And finally, a 'Sigmoid' function was used in the last layer for final outputs. We compiled the model with 'Adam' optimizer and 'binary cross-entropy' loss.

We have also used the standard CNN architecture of VGG16 to classify images under the category of cyclone or non-cyclone. VGG16 is a Convolutional Neural Network that is easy to understand and implement. It has 16 layers, including convolution and pooling layers that are placed on top of each other.

Support Vector Machines (SVM) : SVMs are primarily utilized in classification and regression tasks. It works by drawing outlines between two classes. In this proposed architecture, we have used convolutional layers to extract the features, and SVM is used for the final classification. The convolutional layer contains 16, 32, 64, 128 filters with the size of 3×3 and a stride of 2, respectively. Then fully connected layers have been added with 256, 128 units, respectively. Finally, the last output layer was an SVM Classifier. In the feature extraction layers, ReLU has been used as an activation function, and for the SVM classifier, a Linear activation function with 'l2' regularization has been used. We have compiled the model with the 'Adam' optimizer and loss function 'Hinge'.

K- Nearest Neighbor (KNN) : KNN is based on the idea that every data point that is close to another belongs to the same class. It selects a number k as the closest neighbor to the data point to be identified. We have used convolutional layers for extracting features and used them as input to the KNN classifier to predict the class. This model has been recently used for image classification purposes [22]. For the implementation of the KNN classification, we utilized the Python Scikit-learn toolkit.

Random Forest (RF) : A random forest is a collection of decision trees. This ensemble technique works well for classification and regression purposes.

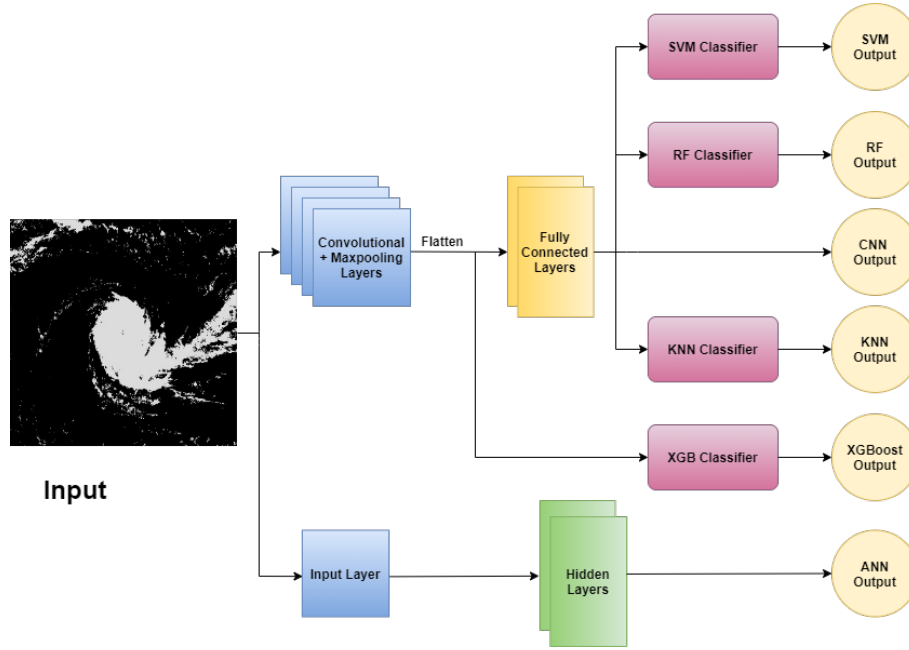


Fig. 2. Proposed Architecture of Different Models.

It uses majority voting for final classification. Because of the local minimum, vanishing gradient, and overfitting difficulties, classical CNNs can face a tough time achieving the optimum generalization [23]. We have used convolutional layers as feature extractors and used them as input to Random forest. By doing so, our model is more generalized and reduces the possibility of overfitting. We have used 50 trees in the random forest.

Gradient Boosting (XGBoost) : The boosting approach is based on the ensemble method's idea. We have used an extreme gradient boosting (XGBoost) classifier, a tree-based algorithm for classification with a custom loss function. We have built a feature extractor and used the features as the input to the XGBclassifier.

Artificial Neural Network (ANN) : ANN has been recently used in many satellite image classification problems [15],[24]. We have built an ANN model having two hidden layers with 128 and 64 units, respectively, and Rectified Linear Unit (ReLU) as an activation function. The last layer is an output layer with an activation function, 'Sigmoid'. The classifier has been compiled with the 'Adam' optimizer and the 'binary cross-entropy' loss function.

3.4 Model for Cyclone Eye Localization

You Only Look Once (YOLO) : It is an object detection algorithm that produces extremely accurate findings in a short time. We have used the YOLOv4 version for detecting a cyclone’s eye or the center rotating area. Fast region-based CNN (R-CNN) can mistake the background patches as an object [25], wherein YOLOv4, the background error can be very much reduced [26].

YOLOv4 backbone architecture is composed of three parts, CSPDarknet-53, Bag of freebies, Bag of special [27]. We have scaled the original image to (300x300) pixels and used it as input to our network. CSPDarknet-53 network is constructed, which is a series of convolutional and residual blocks. ‘SPP’ block was added over the CSPDarknet, ‘PAN’ used as the neck and a YOLOv3 head completes the architecture of YOLOv4. For training and validating our YOLOv4 model, we employed the Google Colab platform, which provided us with 12GB Nvidia Tesla K18 GPU RAM. However, we have annotated our dataset in the local machine. The architecture of our YOLOv4 model is shown in Figure 3.

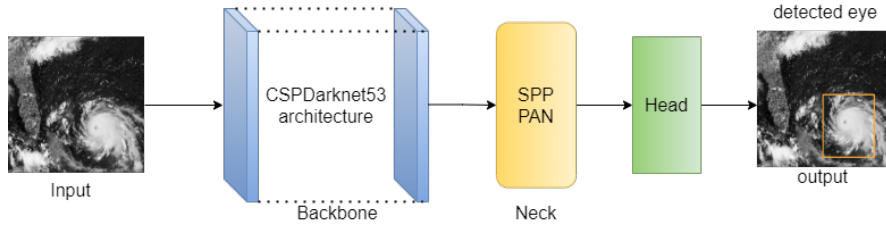


Fig. 3. YOLOv4 Architecture for Cyclone Eye Localization.

4 Experimental Results

We have developed several Deep Learning models to determine whether a satellite image contains a cyclone or not and an object detection model to locate the cyclone’s center rotating area. The detection results of the various models will be shown first, followed by the detection result of the cyclone’s eye.

4.1 Model Evaluation and Validation for Cyclone Detection

We have trained our cyclone detection models for several epochs using seven distinct classifiers with 1151 images. After a few epochs of running, the decreasing behavior of training loss and enhanced accuracy indicated that models have been well-trained on the dataset. The models are evaluated with 309 images, and we have displayed the results in Table 1.

Table 1. Performance Summary for Various Classifiers.

Model	F1-Score	Accuracy
CNN	0.95	95.15%
SVM	0.94	94.17%
KNN	0.86	86.73%
RF	0.93	93.20%
XGBoost	0.91	91.26%
ANN	0.88	88.67%
VGG16	0.93	93.20%

From Table 1, we can see that we have reached over 95% accuracy for CNN, 94% for SVM, 93% for RF, 93% for VGG16, 91% for XGBoost, 88% for ANN, and 86% with our KNN model. Using the convolutional layer as a feature extractor and classifying the class with different classifiers helped the models to improve the accuracy; the effectiveness of preprocessing can also be noticed.

We have calculated the accuracy by dividing the total number of estimates by the number of right estimates. We have also calculated Precision(P), Recall(R), and F1-measure as evaluation metrics.

Table 2. Accuracy Based on Cross Validation.

Model	K-fold	Stratified K-fold
CNN	98.29%	98.42%
SVM	98.84%	99.66%
ANN	96.51%	99.52%
VGG16	93.15%	94.73%

We have verified our model’s performance by applying 5 fold ‘Stratified K-fold’ and ‘K-fold’ cross-validation techniques to prevent overfitting. We have achieved 98.42% and 98.23% accuracy with K-fold and Stratified K-fold, respectively, for the CNN model. For SVM, we have reached 99.66%, 98.84% accuracy, respectively. We have also obtained improved performance for other models. Table 2 summarizes the cross-validation effects.

Following an analysis of the overall results, we have decided that the CNN model could be utilized as a benchmark model for cyclone prediction, using satellite images. In Figure 4, a bar chart depicts the comparison among the mentioned models based on metrics evaluation.

4.2 Model Evaluation and Validation for Cyclone Eye Detection

For predicting the cyclone’s center rotating area, 837 images have been manually annotated for training and testing the YOLOv4 model. The model has been trained on Google Colab for 6000 iterations. After several iterations, the training

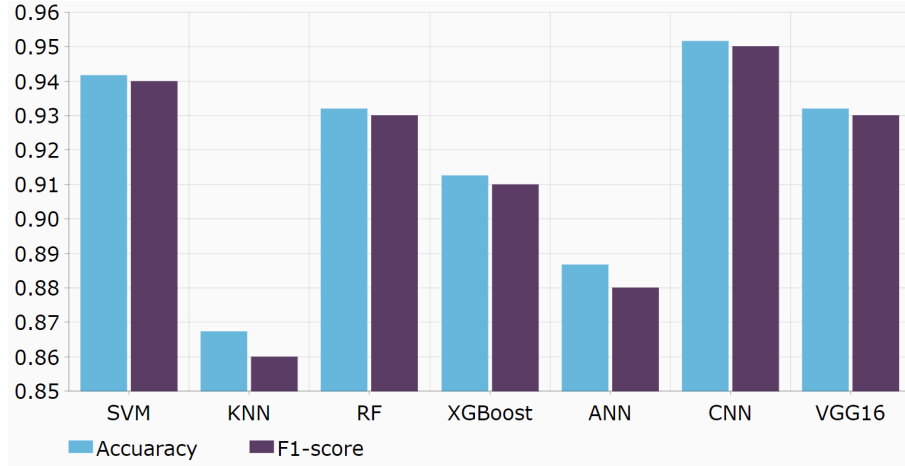


Fig. 4. Performance Comparison Graph for All The Classifiers.

loss started to decrease. We reached a training loss of average 0.5723 with a learning rate of 0.00001.

The performance of our model has been evaluated on 188 test images. The anticipated bounding box of an image is compared to the labeled bounding box of the ground truth image. To test the model's performance, we have used precision, recall, 'mAP', and 'IoU' as evaluation metrics. Table 3 shows the validation results of cyclone eye localization.

Table 3. Performance Evaluation Metrics for YOLOv4.

Model Name	Recall	F1-Score	Average IoU	mAP(mean average precision)
YOLOv4	0.92	0.89	65.37%	87.86%

The ratio of the intersection and union of the anticipated boundary and the ground truth (real cyclone eye boundary) is denoted by 'IoU.' A cyclone eye prediction is considered True Positive (TP) if $\text{IoU} \geq 0.5$, False Positive (FP) if $\text{IoU} < 0.5$. The mAP score indicates a model's performance; the greater the score, the better the prediction. We have reached an mAP of 87.86% for an average IoU of 65.37%. From Table 3, we can also see the Recall and F1-Score of our model are 0.92 and 0.89, respectively.

After evaluating the model's performance, we can observe that it is accurate in detecting a cyclone's eye. Even if a satellite image contains multiple cyclone eyes, the model can identify them effectively.

5 Conclusion & Future Works

Considering cyclones are one of the most destructive natural calamities, automated cyclone prediction is constantly in high demand. We have offered a solution to this major challenge using satellite images and Deep Learning principles. To efficiently detect a cyclone and its core cloud pattern, we have developed a two-step methodology. Although finding a benchmark dataset was a barrier to our accomplished task, our applied technique generated significant outcomes. However, we would want to broaden our future work by using additional data (wind speed, surface temperature, air pressure, etc.) for forecasting and tracking cyclones, and predicting a more accurate center position. Overall, the research reveals that the suggested Deep Learning models could be employed for cyclone prediction with more prominent accuracy and speed than the existing works.

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